

Which kit do you have?

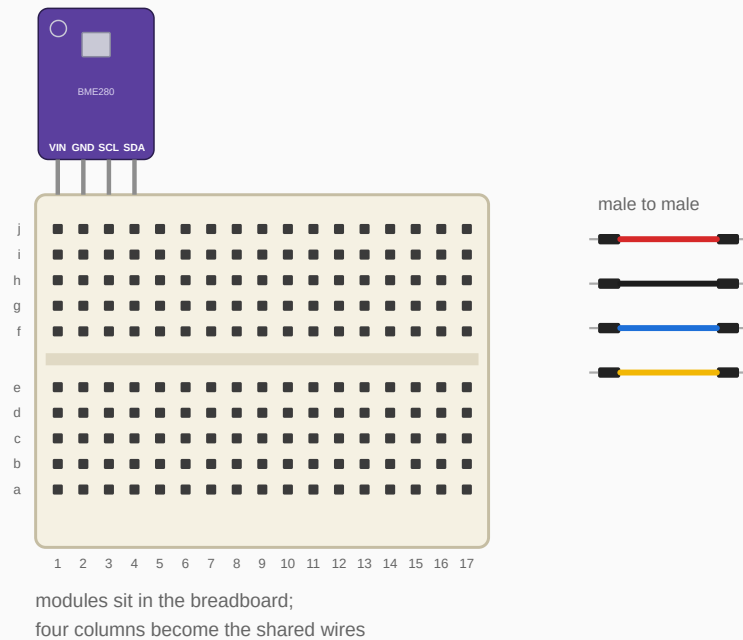
Two builds, one station. The difference is one part: a breadboard. Check the bag, pick a column, and follow only that kit's pages from here.

Kit A

Kit A: breadboard

A small plastic slab full of holes is in the bag.
Sensor and screen run at the same time, joined through the breadboard. Twelve jumper wires with a pin at both ends (male to male).

This is the US workshop kit.
About 25 minutes to a working station.



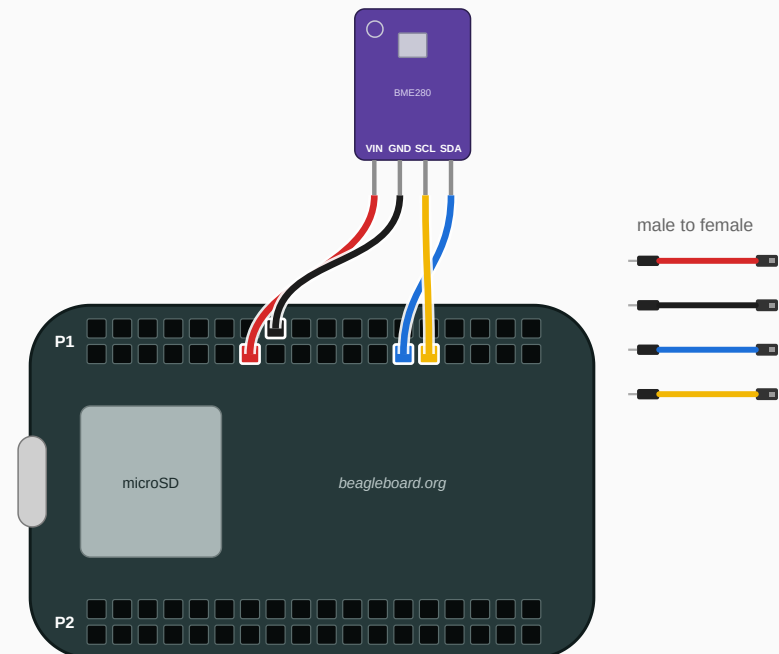
Kit B

Kit B: direct, no breadboard

No breadboard in the bag.
Four wires go straight from the board's sockets to one module. Sensor first; then the screen, on its own. The wires need a pin at one end and a socket at the other (male to female).

This is the UK bench until a breadboard arrives,
and it works with any kit. About 15 minutes.

one module at a time, wired straight from the board's sockets

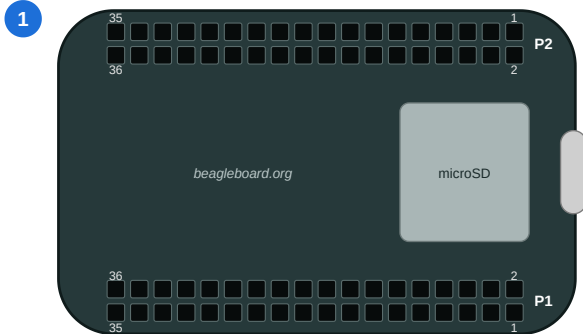


Same board, same four pins, same software, same checks. Only the joining differs. A Kit B bench becomes a Kit A bench the day a breadboard turns up.

Kit A: what is in the bag

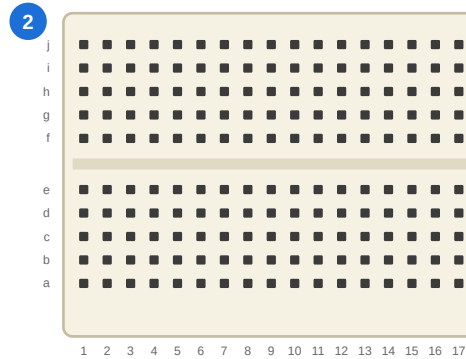
Kit A: breadboard

Tip everything out and match each thing to a picture. Everything is drawn at the same scale as every later picture.



PocketBeagle 2

The computer. Two black strips of sockets (P1, P2) along the long edges, tiny numbers at the strip ends. Hold it by the edges. The side with the microSD slot and the dog logo is the side you use.



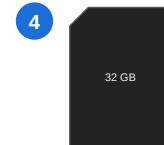
Mini breadboard, 170 holes

A small plastic slab, 17 columns by 10 rows, with a groove across the middle. No red or blue power lines on this size; that is expected.



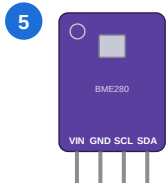
USB-C cable

Must be a data cable. A charge-only cable looks identical and is the most common reason nothing works.



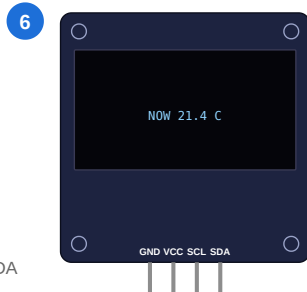
microSD card

8 GB or more. Holds the whole operating system.



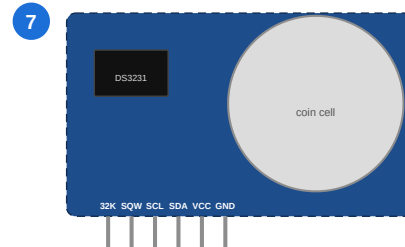
BME280 sensor

Fingernail sized, purple or blue, four pins (a few have six): VIN or VCC, GND, SCL, SDA in tiny letters. The small metal can is the sensor itself. If the pins are loose in the bag instead of soldered on, stop: that is a different job.



OLED screen (SSD1306, 0.96 inch)

A square board with a dark glass window and four pins on one edge: VCC, GND, SCL, SDA. The order of VCC and GND **differs between makers. Read yours now and write it down.** Getting those two backwards is the one mistake in this guide that kills a part instantly.



DS3231 clock (optional)

The one with the round coin cell holder. Six pins; only SCL, SDA, VCC, GND are used. Leave it in the bag on your first build. No battery until you have read the battery note in [HARDWARE.md](#).

8 Jumper wires, male to male

A pin at both ends. 12 needed, 16 with the clock.



Any four colours work if you use them the same way throughout.

Not in the bag and not needed: a monitor, keyboard or power supply for the board. Your laptop is all three, through the USB-C cable.

Kit A: lay it out like this before you start

Kit A: breadboard

This is the arrangement every later picture uses. You sit at the bottom edge. Cable and card stay where they are until Parts 3 and 4.

1 Modules, pins toward you

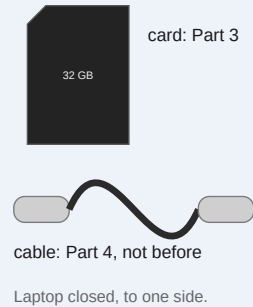


2 Jumpers in four rows, by colour



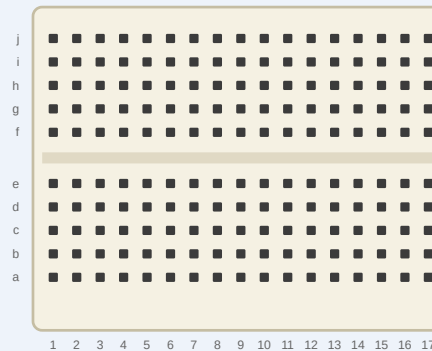
Count them now. Hunting for a missing wire at step 3 costs ten minutes of doubt.

3 Not yet



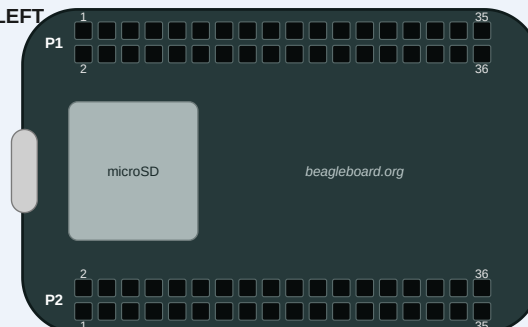
4 Breadboard, the long way across, 4 cm below the modules

Column numbers run left to right, 1 to 17. The groove across the middle splits each column into a top half (rows f to j) and a bottom half (rows a to e). They are not joined to each other.



Modules go along the top edge with their pins in row j. Four columns in the middle, 5 to 8, become the shared wires. Nothing goes in the bottom half in this build.

5 PocketBeagle 2, sockets up, USB-C port to the LEFT



Turned so P1 is the strip nearest the breadboard. Pin 1 is then at the left end of P1, by the USB-C port.

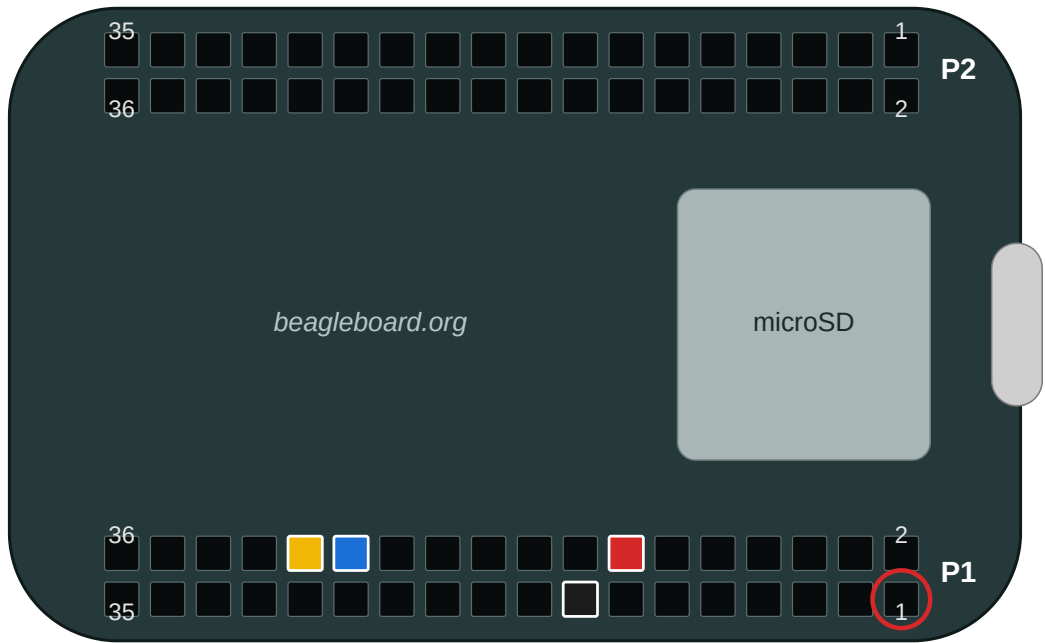
Find the tiny printed 1 and 2 there before going on.

Every wire that leaves P1 goes straight up into the breadboard.

your table, seen from above

Where the four pins are

Board held with the sockets facing you and the USB-C port on the right, as BeagleBoard photographs it. On your table it is turned round; the numbers are the same.



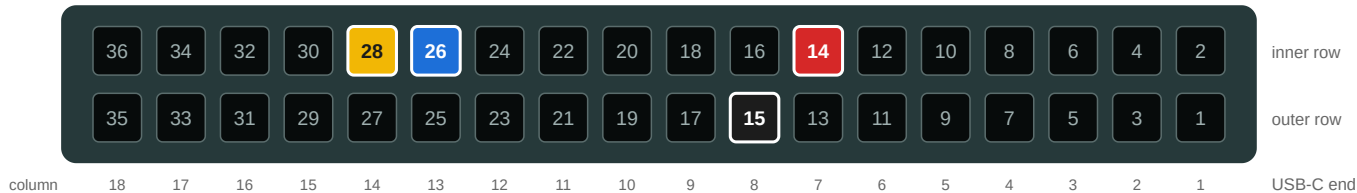
P1.1 is 5 V. Nothing ever goes in here.

The four we use

- 14 P1.14 VDD_3V3**
3.3 V out. column 7, inner row, from the USB-C end.
- 15 P1.15 GND**
ground. column 8, outer row, from the USB-C end.
- 26 P1.26 I2C2_SDA**
SDA, data. column 13, inner row, from the USB-C end.
- 28 P1.28 I2C2_SCL**
SCL, clock. column 14, inner row, from the USB-C end.

Trust the numbers printed on the board over this picture. Find the printed 1 and 2 at the USB-C end of P1 and the 35 and 36 at the far end. If they are there, the columns are right.

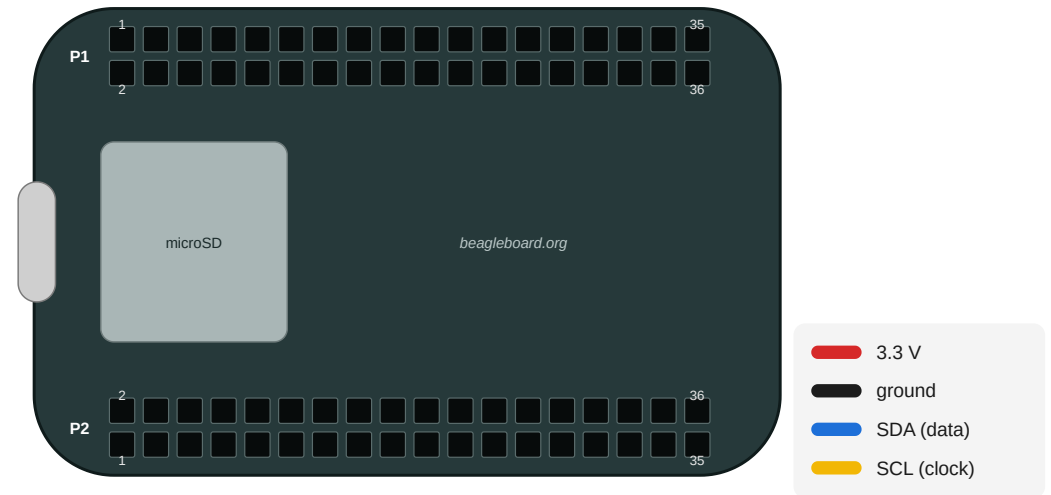
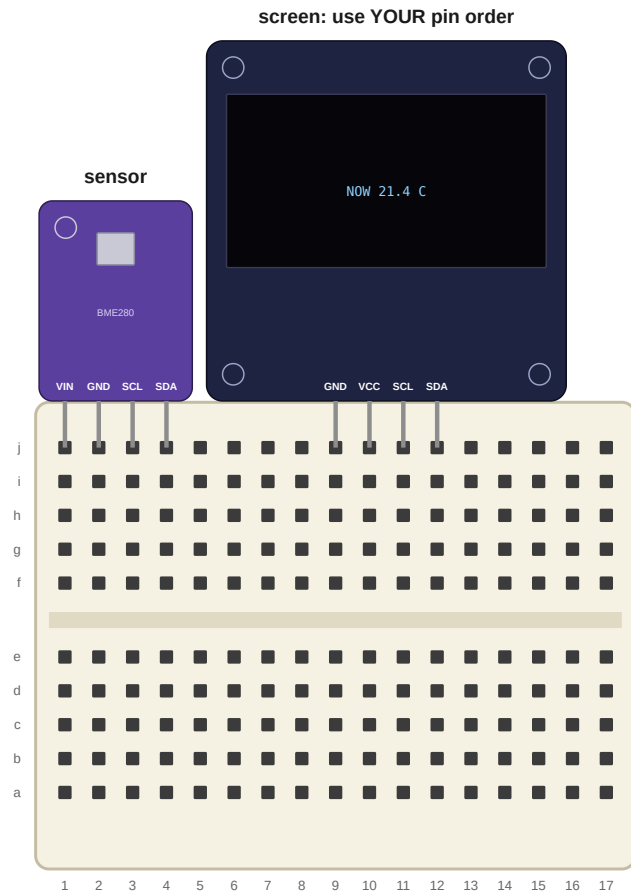
P1 close up. Column 1 is at the USB-C end. Outer row = odd pins, inner row = even pins.



Kit A, step 1 of 4: seat the modules

Kit A: breadboard

Sensor pins in row j, columns 1 to 4. Screen pins in row j, columns 9 to 12. Bodies hang off the top edge. USB cable still in the bag.



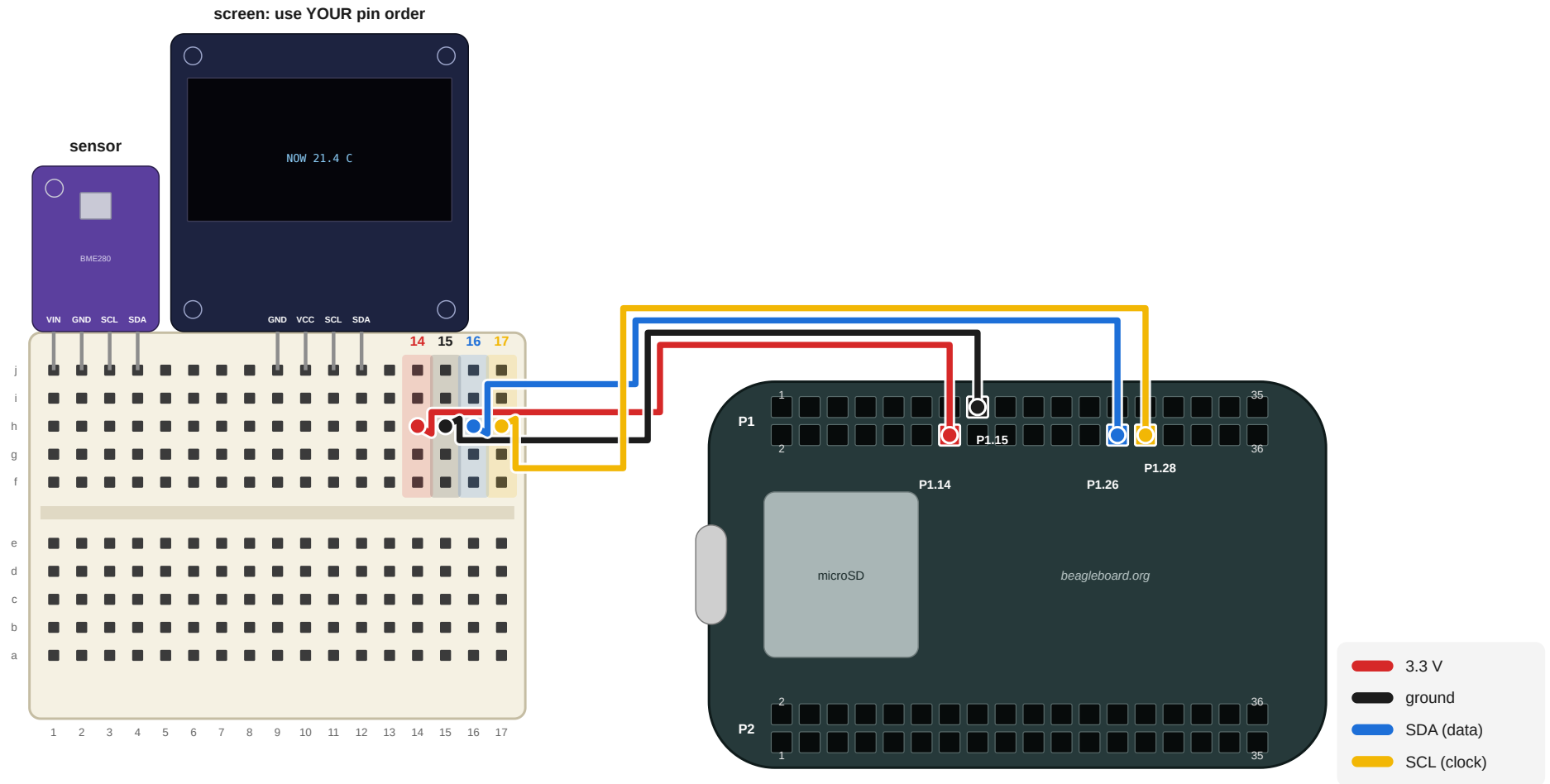
board turned so P1 is its top strip, pin 1 nearest the breadboard, USB-C on the left.

Press each module in firmly and evenly; it takes more force than you expect. Pins in row j, one per column. Nothing goes in the bottom half of the board. The bodies hang off the top edge. Columns 14 to 17, nearest the computer, stay empty: they become the four shared wires in step 2.

Kit A, step 2 of 4: four wires from the board

Kit A: breadboard

The four columns nearest the board become the shared wires: 14 is 3.3 V, 15 is ground, 16 is SDA, 17 is SCL. Red P1.14 to 14, black P1.15 to 15, blue P1.26 to 16, yellow P1.28 to 17.



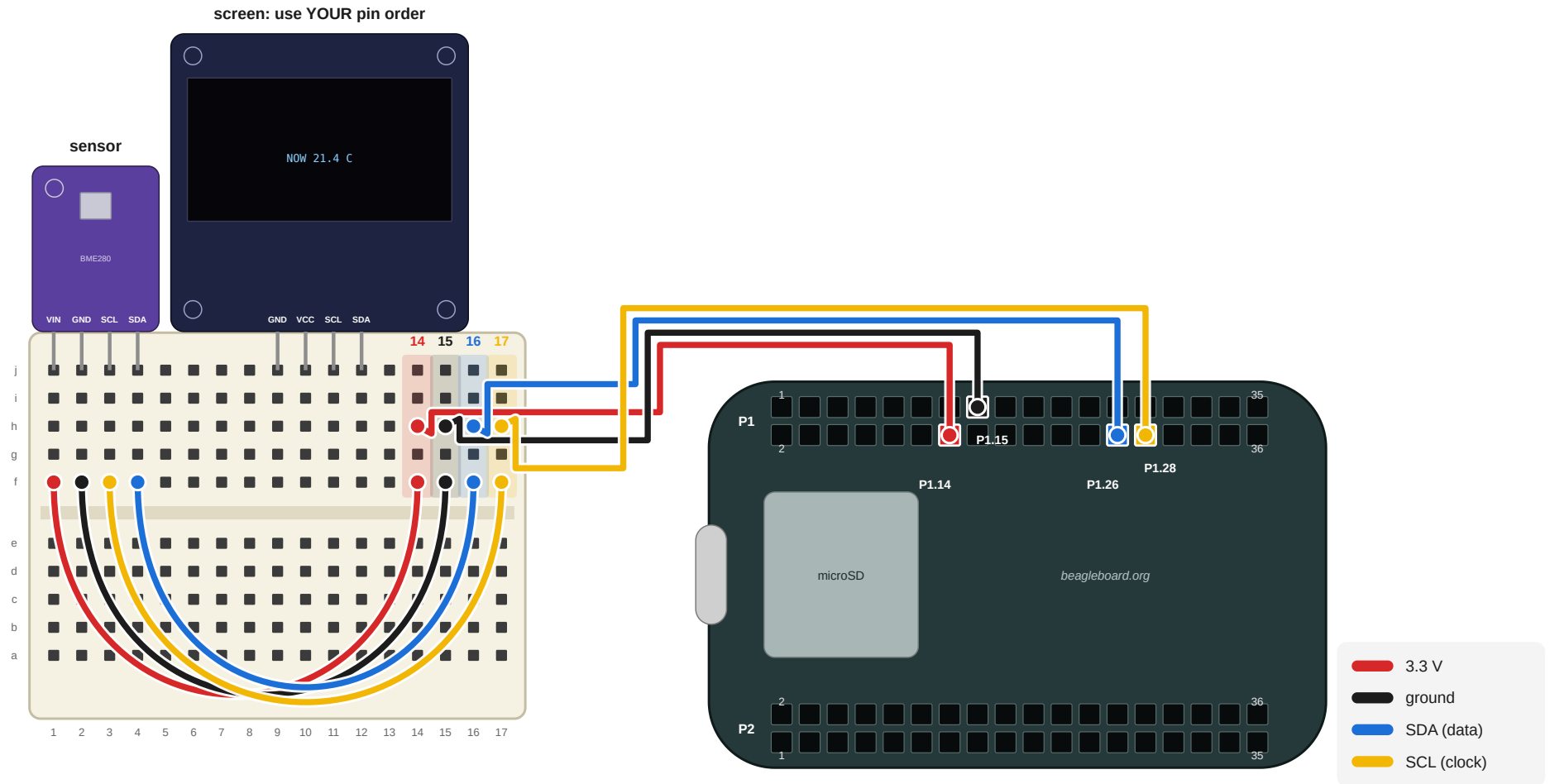
board turned so P1 is its top strip, pin 1 nearest the breadboard, USB-C on the left.

Each shared column is one wire inside the breadboard, five holes long. The board's wire, the sensor's wire and the screen's wire will each use one hole of it. Say it out loud: red leaves 14 and arrives at column 14. Black leaves 15 and arrives at 15. Blue leaves 26 and arrives at 16. Yellow leaves 28 and arrives at 17.

Kit A, step 3 of 4: the sensor's four wires

Kit A: breadboard

From the sensor's columns to the shared columns, all in row f: VIN to 14, GND to 15, SDA to 16, SCL to 17. Read the sensor's letters before each wire.



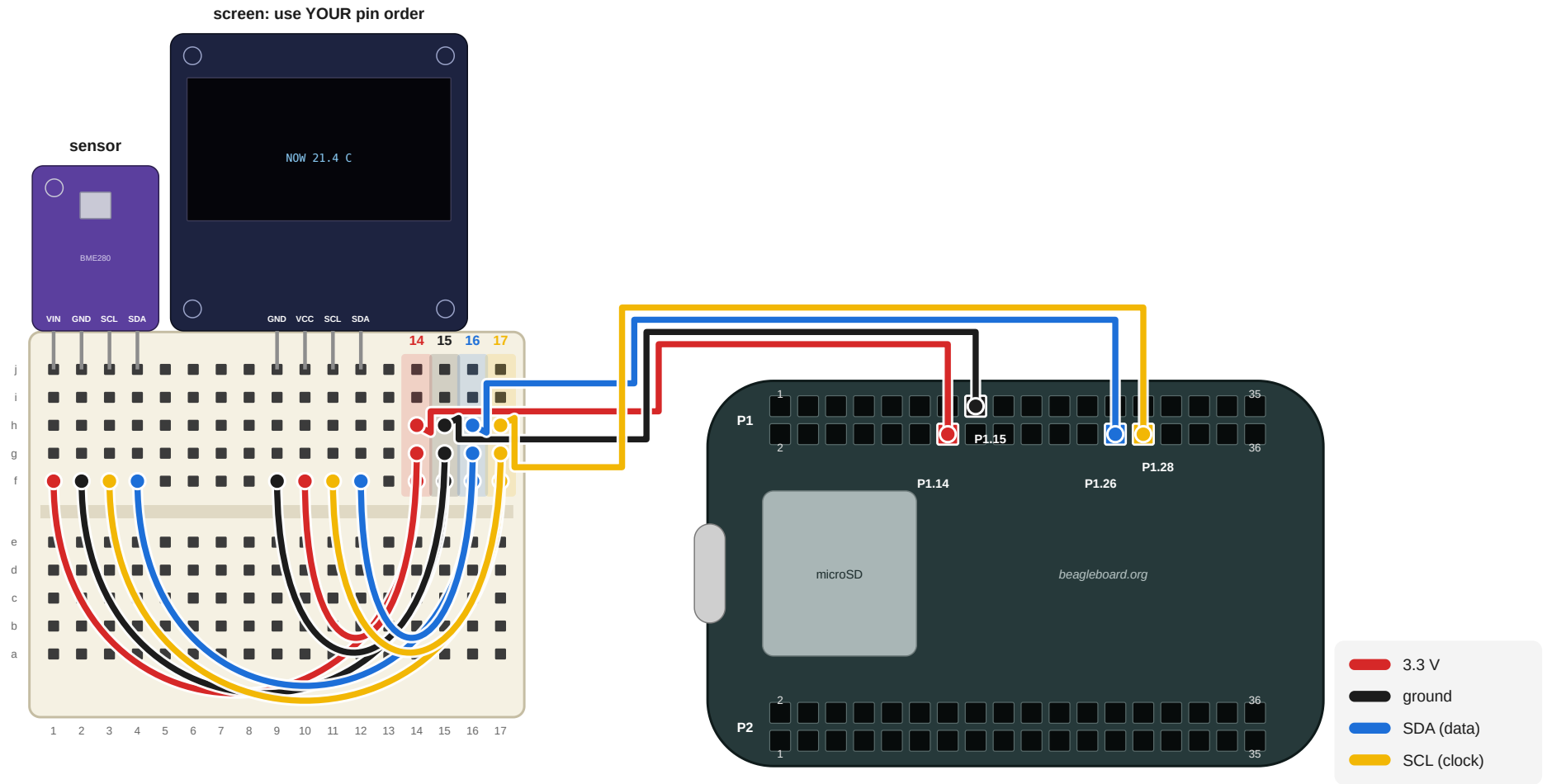
board turned so P1 is its top strip, pin 1 nearest the breadboard, USB-C on the left.

Sensor wires start in row f under each sensor pin and end in row f of a shared column. They swing across the empty bottom half; they touch the board only at their ends. Follow each one with a finger back to the sensor and read the letter at that column: red must end at VIN, black at GND. VIN and GND swapped kills the sensor at power on.

Kit A, step 4 of 4: the screen's four wires

Kit A: breadboard

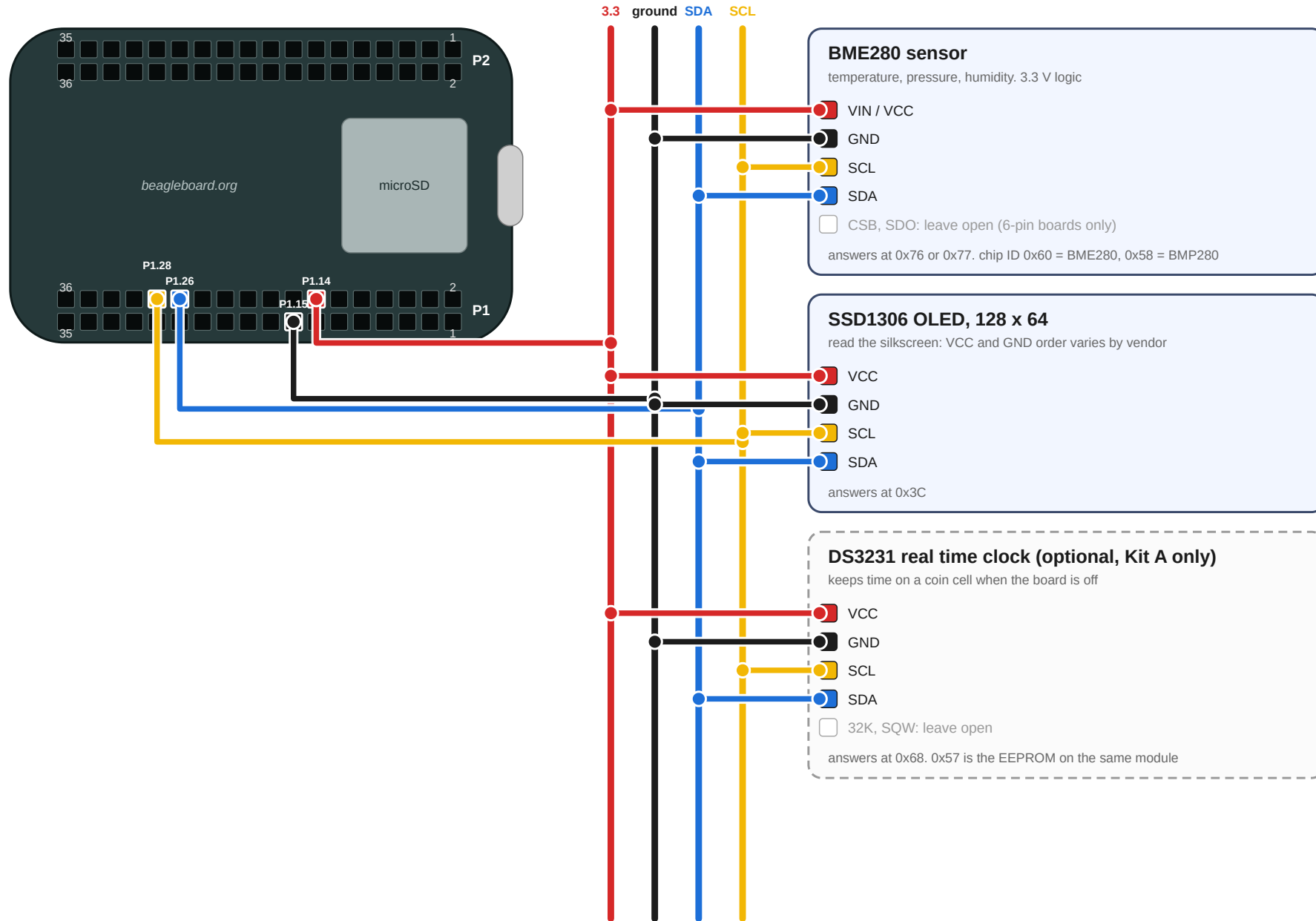
Same for the screen, in YOUR screen's pin order, ending in row g: VCC to 14, GND to 15, SDA to 16, SCL to 17. Then check every wire out loud.



Screen wires start in row f under each screen pin and end in row g of a shared column, in the pin order you wrote down for YOUR screen. Twelve wires in total. Nothing in P1.1. No bare pins touching. That is the whole circuit.

The four wires, as a schematic

The same four wires for every module: power, ground, data, clock. This is the whole circuit, in either kit.



Kit A: four breadboard columns. Kit B: four wires, one module